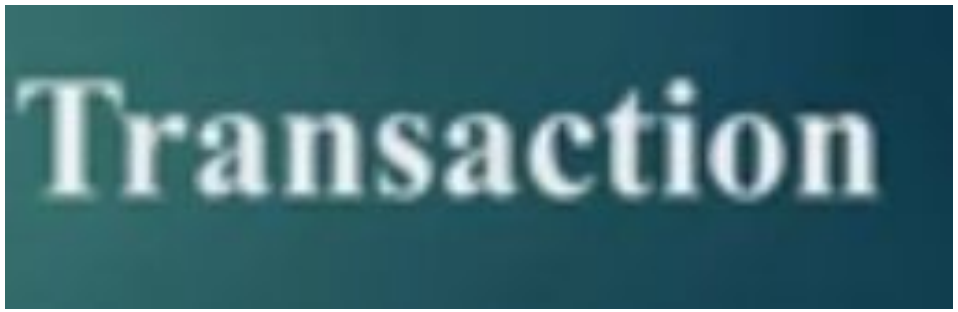

Transaction

Introduction

- ✔ Transactions are a set of operations used to perform a logical set of work. It is the bundle of all the instructions of a logical operation. A transaction usually means that the data in the database has changed. One of the major uses of DBMS is to protect the user's data from system failures. It is done by ensuring that all the data is restored to a consistent state when the computer is restarted after a crash. The transaction is any one execution of the user program in a DBMS. One of the important properties of the transaction is that it contains a finite number of steps. Executing the same program multiple times will generate multiple transactions.

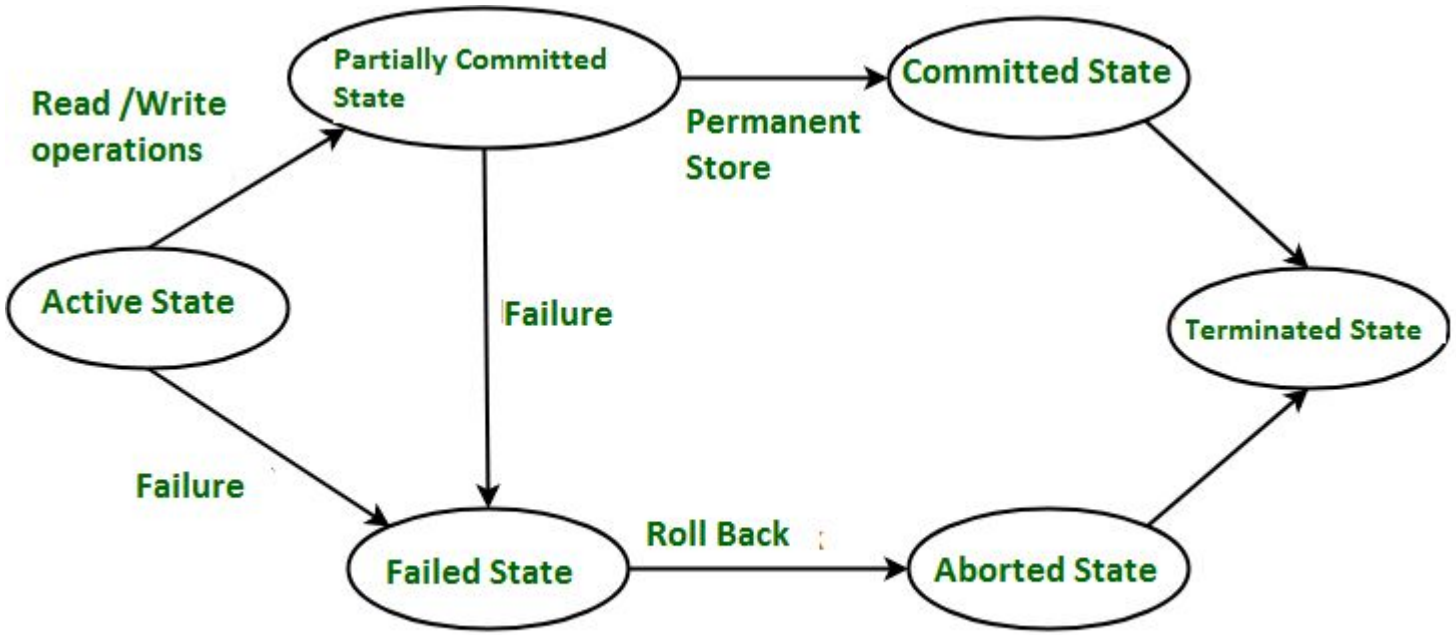
Transactions

- A transaction is a program including a collection of database operations, executed as a logical unit of data processing.
- The operations performed in a transaction include one or more of database operations like insert, delete, update or retrieve data.
- It is an atomic process that is either performed into completion entirely or is not performed at all.
- A transaction involving only data retrieval without any data update is called read-only transaction.

Transaction

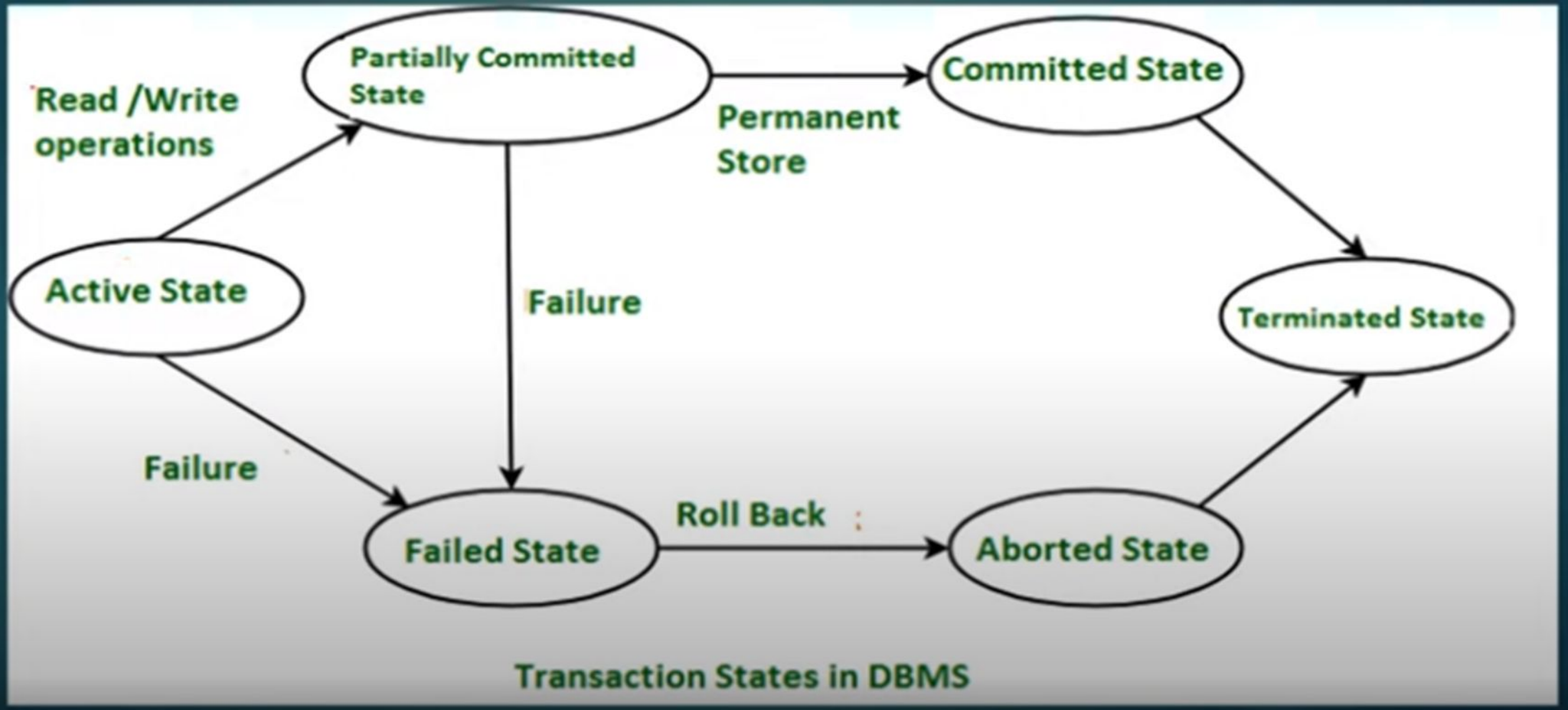
- Transaction States,
  - Active state
  - Partially committed state
  - Committed state
  - Failed state
  - Aborted state
  - Terminated state

Introduction



Transaction States in DBMS

Transaction



Transaction

- The transaction has the four properties,
  - Atomicity
  - Consistency
  - Isolation
  - Durability

Transaction

- Atomicity
  - Each transaction is treated as one unit and either run to completion or is not executed at all.
- Transaction T consisting of T1 and T2. A consists of Rs. 500 and B consists of Rs. 100

T1	T2
Read(A) A:= A-100 Write(A)	Read(B) Y:= Y+100 Write(B)

Transaction

- **Consistency** – The database must remain in a consistent state before and after any transaction.
- Durability – If a committed transaction brings about a change, that change should be durable in the database and not lost in case of any failure.

Transaction

- Isolation- A transaction should be executed as if it is the only one in the system. There should not be any interference from the other concurrent transactions that are simultaneously running.
- Transaction T1 is being executed and using the data item A, then that data item can't be accessed by any other transaction T2 until the transaction T1 ends.